MI Net Growth Levels Management - Change Summary

Developer: Sh3hann | **Branch:** main / feature/mi-net-growth-levels | **Date:** November 2026

Repository: https://github.com/Sh3hann/mi-net-growth.git

Implemented a full-stack system to manage MI Net Growth Levels for the SIA (Sales Incentive Automation) system using PostgreSQL, Django REST Framework, and React.js with full CRUD functionality, sequential NET_ID auto-generation, range contiguity/overlap prevention, file attachment uploads, and Role-Based Access Control (CFO, DGM, RM).

No	File	Path	Used In	What Changed
1	**models.py**	`backend/growth_levels/models.py`	Database Table & ORM	Created SIA_MI_NET_GROWTH_LEVELS table with sequential NET_ID auto-generator, range validation, and overlap checks.
2	**views.py**	`backend/growth_levels/views.py`	REST API & Business Logic	Built MiNetGrowthLevelViewSet supporting CRUD, statistics, file download, and RBAC enforcement (CFO/DGM/RM).
3	**urls.py**	`backend/growth_levels/urls.py`	Backend Routing	Mapped /api/mi-net-growth-levels/ routes using DRF DefaultRouter for all endpoints.
4	**serializers.py**	`backend/growth_levels/serializers.py`	API Serialization	Created MiNetGrowthLevelSerializer with decimal validation, overlap checks, attachment URL builders, and audit user headers.
5	**pagination.py**	`backend/growth_levels/pagination.py`	Data Pagination	StandardResultsSetPagination supporting customizable page size and unpaginated query parameters.
6	**settings.py**	`backend/growth_backend/settings.py`	Backend Settings	Configured PostgreSQL with SQLite fallback, CORS headers, Media root and URL for file storage.
7	**tests.py**	`backend/growth_levels/tests.py`	Automated Testing	Added 11 unit tests covering CRUD, range overlap rejection (400), percentage limits (400), and RBAC permissions (403).
8	**seed_data.py**	`backend/seed_data.py`	Data Seeding	Populated initial enterprise growth tiers (NET001 to NET006) with realistic ranges, percentages, and sample attachments.
9	**docker-compose.yml**	`backend/docker-compose.yml`	PostgreSQL Setup	Containerized PostgreSQL 15 alpine database service listening on port 5432.
10	**api.js**	`frontend/src/services/api.js`	API Client Service	Axios client with multipart/form-data support, dynamic role headers, error message parser, and blob download.
11	**Navbar.jsx**	`frontend/src/components/Navbar.jsx`	Top Navigation	Enterprise navigation with title, simulator modal trigger, and live RBAC role switcher (CFO, DGM, RM).
12	**StatsCards.jsx**	`frontend/src/components/StatsCards.jsx`	Dashboard KPIs	Metric cards displaying Total Levels, Active Levels, Inactive Levels, Average Incentive %, and Max Range Ceiling.
13	**RangeVisualizer.jsx**	`frontend/src/components/RangeVisualizer.jsx`	Spectrum Analyzer	Visual range coverage bar confirming continuous non-overlapping active tiers from $0 to maximum ceiling.
14	**FilterBar.jsx**	`frontend/src/components/FilterBar.jsx`	Toolbar & Actions	Search by level or NET_ID, status filter tabs (All, Active, Inactive), CSV export button, and Add Level button.
15	**GrowthLevelTable.jsx**	`frontend/src/components/GrowthLevelTable.jsx`	Data Table	Table showing NET_ID, ranges, incentive %, active status pill, file download links, and role-based action buttons.

models.py

No	Change	Details
1	**Table Name**	Created SIA_MI_NET_GROWTH_LEVELS table mapped through Django ORM db_table setting.
2	**NET_ID Field**	CharField unique=True (max_length=20) with automated sequence generator (NET001, NET002...).
3	**Range Fields**	LOWER_RANGE and UPPER_RANGE DecimalField(14,2) with non-negative and upper > lower constraints.
4	**Overlap Validation**	Model clean() verifies that no two Active records overlap in the lower_range to upper_range interval.

views.py (growth_levels)

No	Change	Details
1	**CRUD Endpoints**	MiNetGrowthLevelViewSet provides POST (create), GET (list), GET // (retrieve), PUT/PATCH (update), DELETE (destroy).
2	**Search & Filter**	Implemented ?search= (Level, NET_ID, Comments) and ?status=Active|Inactive query filtering.
3	**Statistics API**	Added /api/mi-net-growth-levels/statistics/ action calculating total, active, inactive, avg %, and max ceiling.
4	**RBAC Permissions**	check_role_permissions() blocks RM from all writes (403) and blocks DGM from delete actions (403).

views.py (growth_levels) - Continued

No	Change	Details
5	**Download Attachment**	/api/mi-net-growth-levels//download_attachment/ streams binary file with Content-Disposition header.
6	**Multipart Support**	parser_classes = [MultiPartParser, FormParser, JSONParser] allows file uploads and JSON interchangeably.
7	**Ordering Logic**	Supported ?ordering= query parameter to dynamically sort by lower_range, percentage, or created_date.
8	**Unpaginated Export**	Supported ?paginate=false to return all records for reporting, visualizations, and client calculations.

urls.py

No	Change	Details
1	**DefaultRouter Route**	router.register(r'mi-net-growth-levels', MiNetGrowthLevelViewSet) creates full REST URL tree.
2	**API Integration**	Mounted in growth_backend/urls.py under path('api/', include('growth_levels.urls')).
3	**Media File Handler**	Added static(settings.MEDIA_URL, document_root=settings.MEDIA_ROOT) to serve uploads in dev.

serializers.py

No	Change	Details
1	**Key Normalization**	to_internal_value() maps incoming uppercase keys (LEVEL, LOWER_RANGE) to lowercase model fields.
2	**Range Overlap Check**	Validates range bounds against database active records and returns clear 400 Bad Request if conflicts exist.
3	**Percentage Boundaries**	Enforces 0.00% <= percentage <= 100.00% returning field error if boundary is violated.
4	**Audit Header Tagging**	create() and update() auto-populate created_user and updated_user from request X-User-Role / X-User-Name.
5	**Attachment Metadata**	SerializerMethodFields dynamically expose attachment_name and absolute attachment_url.

pagination.py

No	Change	Details
1	**Standard Pagination**	StandardResultsSetPagination with page_size=10, page_size_query_param='page_size', max_page_size=100.
2	**Metadata Response**	Returns count, total_pages, current_page, next, and previous links alongside data results.

docker-compose.yml

No	Change	Details
1	**PostgreSQL 15 Container**	Configured postgres:15-alpine image with POSTGRES_DB=mi_net_growth_db on port 5432.
2	**Volume Persistence**	Mapped postgres_data volume to /var/lib/postgresql/data for local data durability.

api.js (Frontend Service)

No	Change	Details
1	**Axios Client**	Created centralized Axios instance with baseURL (http://localhost:8000/api) and timeout.
2	**RBAC Interceptors**	Attaches current active role (X-User-Role) and user name from localStorage to every outbound request.
3	**Multipart Form Support**	growthLevelsApi.create() and update() automatically inspect data and apply multipart/form-data.
4	**Blob Download Helper**	downloadAttachment() requests responseType: 'blob' and generates temporary DOM anchor to download.
5	**DRF Error Formatter**	formatApiError() unpacks DRF error dictionaries into human-readable toast notification strings.

Frontend Components (React.js)

No	Change	Details
1	**Navbar.jsx**	Displays system title, simulator modal shortcut, and live role switcher (CFO, DGM, RM).
2	**StatsCards.jsx**	5 clean metrics cards showing Total Levels, Active Levels, Inactive Levels, Avg Incentive %, Max Range.
3	**RangeVisualizer.jsx**	Visual tier coverage bar displaying continuous non-overlapping distribution from $0 to ceiling.
4	**FilterBar.jsx**	Search by level or NET_ID, status filter tabs (All, Active, Inactive), CSV export, and Add Level button.
5	**GrowthLevelTable.jsx**	Table displaying NET_ID, Level, Ranges, Incentive %, Active status pill, Attachment download, and actions.
6	**LevelFormModal.jsx**	Modal with client-side range checks, real-time overlap collision alert, and file upload input.
7	**DeleteConfirmModal.jsx**	Confirmation dialog showing tier details before deletion, restricted to CFO.
8	**TierDetailDrawer.jsx**	Slide-over drawer displaying full tier specs, payout scenarios, audit trail, and document downloader.
9	**Toast.jsx**	Toast notification container providing real-time feedback for create, update, delete, and validation errors.

SimulatorModal.jsx

No	Change	Details
1	**Deal Volume Input**	Allows entering gross net growth amount with quick presets ($25k, $100k, $250k, $500k, $1M).
2	**Bracket Matching**	Evaluates active tiers to find matching qualification tier and applicable incentive percentage.
3	**Formula Breakdown**	Displays exact formula calculation and distance needed to unlock the next higher tier bracket.

.env (Backend & Frontend)

No	Change	Details
1	**USE_POSTGRES**	USE_POSTGRES=False (defaults to SQLite for instant local dev) or True to connect to PostgreSQL.
2	**DB Connection Keys**	DB_NAME=mi_net_growth_db, DB_USER=postgres, DB_PASSWORD=postgres, DB_HOST=localhost, DB_PORT=5432.
3	**VITE_API_URL**	VITE_API_URL=http://localhost:8000/api for Vite React client.

User Flow

No	Flow	Steps
1	**View & Search Tiers**	User navigates to dashboard, views KPI metrics and active range spectrum, searches by level name or NET_ID, and filters by status (All, Active, Inactive).
2	**Add New Tier (CFO/DGM)**	User clicks 'Add Level', fills level title, lower & upper ranges, percentage, comments, and optional file. Client and backend validate range contiguity and overlap before committing.
3	**Edit Tier (CFO/DGM)**	User clicks Edit icon on a row, form pre-populates with existing data. User modifies range or replaces attachment. Table and KPI stats auto-refresh upon save.
4	**Delete Tier (CFO Only)**	CFO clicks Delete icon, confirmation dialog displays record summary. Deletion permanently removes tier. DGM and RM users have Delete button disabled (HTTP 403 on API).
5	**Simulate Deal Payout**	User clicks 'Simulator' in header, inputs gross deal volume (e.g. $250,000), engine highlights matching tier, applicable percentage (8.75%), and calculates payout ($21,875.00).

Testing Checklist

No	Test Case	Expected Result
1	**Database Table Creation**	Table SIA_MI_NET_GROWTH_LEVELS created in PostgreSQL via makemigrations and migrate.
2	**Auto NET_ID Generation**	Sequential ID (NET001, NET002, ...) auto-generated on record creation without manual input.
3	**List Records API (GET)**	Returns HTTP 200 OK with paginated list, count, total_pages, and results array.

Testing Checklist - Continued

No	Test Case	Expected Result
4	**Single Record Retrieval (GET)**	Returns HTTP 200 OK with single record details matching requested PK ID.
5	**Multipart File Upload (POST)**	Returns HTTP 201 Created; uploaded file stored in media directory and attachment URL returned.
6	**Range Overlap Validation**	Attempting to create or update an overlapping active range returns HTTP 400 Bad Request with conflicting tier names.
7	**Percentage Limit Validation**	Percentage < 0 or > 100 returns HTTP 400 Bad Request.
8	**Upper > Lower Range Validation**	Upper range <= lower range or negative lower range returns HTTP 400 Bad Request.
9	**RM Read-Only Access (RBAC)**	RM role sending POST, PUT, PATCH, or DELETE receives HTTP 403 Forbidden.
10	**DGM Delete Prevention (RBAC)**	DGM role sending DELETE receives HTTP 403 Forbidden. Can execute POST, PUT, and PATCH.
11	**CFO Full Permissions (RBAC)**	CFO role can create, view, update, and delete growth levels.
12	**Attachment File Download**	GET download_attachment streams file with Content-Disposition header and correct filename.
13	**Frontend Table & Auto-Refresh**	Table renders active pills, formatted currency, and auto-refreshes immediately after mutations.
14	**Automated Backend Test Suite**	All 11 unit tests in tests.py pass successfully (Ran 11 tests in 0.101s, OK).

Git Status & Repository State

Remote Origin: https://github.com/Sh3hann/mi-net-growth.git
Branches Synced: main, feature/mi-net-growth-levels (Up to date with origin)
Author: Sh3hann <akilshehan760@gmail.com>
Commit Log:
• 8831c9e (origin/main, main) chore: sync with GitHub origin main and retain full system documentation
• 7d37abe docs(pdf): generate change summary and git status PDF documentation
• 4e7d294 feat(mi-net-growth): complete full-stack implementation with Django ORM, REST API, React frontend, and Postman suite

Tracked Files in Repository:
• backend/growth_levels/models.py ..... [SIA_MI_NET_GROWTH_LEVELS table model]
• backend/growth_levels/views.py ..... [MiNetGrowthLevelViewSet CRUD & RBAC]
• backend/growth_levels/serializers.py ..... [DRF serializer with overlap validation]
• backend/growth_levels/urls.py ..... [API routes registration]
• backend/growth_levels/pagination.py ..... [Standard pagination handler]
• backend/growth_levels/tests.py ..... [11 unit tests passing]
• backend/growth_backend/settings.py ..... [PostgreSQL / SQLite config & CORS]
• backend/seed_data.py ..... [Database seeder for sample tiers]
• backend/docker-compose.yml ..... [PostgreSQL 15 service]
• frontend/src/services/api.js ..... [Axios CRUD service & multipart handler]
• frontend/src/components/Navbar.jsx ..... [Header & RBAC role switcher]
• frontend/src/components/StatsCards.jsx ..... [Executive KPI cards]
• frontend/src/components/RangeVisualizer.jsx ..... [Contiguity spectrum analyzer]
• frontend/src/components/FilterBar.jsx ..... [Search & status filter toolbar]
• frontend/src/components/GrowthLevelTable.jsx ..... [Data table with action triggers]
• frontend/src/components/LevelFormModal.jsx ..... [Add/Edit modal with overlap check]
• frontend/src/components/DeleteConfirmModal.jsx ..... [Deletion confirmation dialog]
• frontend/src/components/SimulatorModal.jsx ..... [Net revenue payout simulator]
• frontend/src/components/TierDetailDrawer.jsx ..... [Tier specs & dossier slide-over]
• frontend/src/components/Toast.jsx ..... [Toast notification system]
• postman/MI_Net_Growth_Levels.postman_collection.json [Postman collection v2.1.0]
• postman/POSTMAN_GUIDE.md ..... [Complete API testing guide]
• run_all.ps1 ..... [One-click startup PowerShell script]